

TEAM

- **TEAM NAME: Team Glitch**
- **MEMBERS & ROLES:**

Ayush Kumar
Lead: robot & simulation, vision & AI, shared memory, Hindi voice

Soumya Raghuwanshi
Research: Indian hospital data and sources

Simran Kumari Keshri
Testing and pitch: voice tests, demo video

SAATHI

A robot guide that reads Indian hospital crowds

PS-01 · Vision-Based Autonomous Robot · Round 2: working prototype in a two-floor Webots hospital

github.com/shaunmortal/saathi



Web3 India | builder base | EXIDO | hack in hills

PROBLEM STATEMENT: PATIENTS GET LOST IN INDIAN OPDs

14–15k
OPD patients a day at one hospital (AIIMS Delhi)

69.6%
of first-time OPD visitors needed a person to find their way (GMC Bhopal)

98%
of older patients at AIIMS geriatric OPD come alone (2026 preprint)

3 h+
total wait for 58% of 400 patients (RIMS Ranchi, 2026)

ONE PATIENT, ONE X-RAY: TODAY vs WITH SAATHI

TODAY

Enters alone signs in English → **Asks the guard** "seedha jao" → **Wrong corridor** walks back → **Asks a nurse** she leaves her patient → **Reaches Radiology** after a long detour

WITH SAATHI

"Mujhe X-ray karwana hai" in Hindi, at the entrance → **Saathi walks with them** at their pace, through the crowd → **Reaches Radiology** and says so

WHAT IT COSTS, AND WHY A ROBOT

Hours lost, appointments missed
patients, mostly first-time, elderly, low-literacy

Help desks and nurses pulled away
every "X-ray kahan hai?" is a nurse off her patient

A fall in a packed hall is noticed late
nobody is watching the floor

Today's robot planners fail in crowds
they freeze, or cut straight through people



Our OPD hall in Webots: Hindi + English signs, token display, pharmacy queue, animated people

THE GAP

No wayfinding robot found in any Indian OPD (our 2024–26 search). Guide robots exist at airports abroad; Indian OPD crowds are denser.

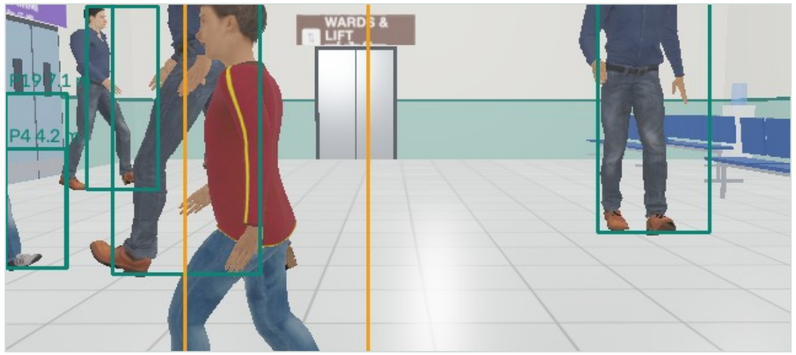
PS-01 · VISION-BASED AUTONOMOUS ROBOT

- AI vision: people + objects
- Drives itself to a goal
- Real-time decisions
- Simulated hospital (allowed)

Sources: AIIMS OPD load, UNI and Free Press Journal, Sep 2026 · GMC Bhopal, IJCMPH 2026 (n=336) · AIIMS geriatric OPD, Research Square preprint, Aug 2026 (n=400) · RIMS Ranchi, Cureus, Sep 2026 (n=400)

SOLUTION: SAATHI, A GUIDE ROBOT THAT READS THE CROWD

Say “Mujhe X-ray karwana hai”. Saathi walks you there at your pace, talks in Hindi, explains every move, and takes the lift when your department is upstairs. All frames below are from our Webots runs.



Sees every person with an ID and distance

YOLO11 + depth camera + 360° LiDAR



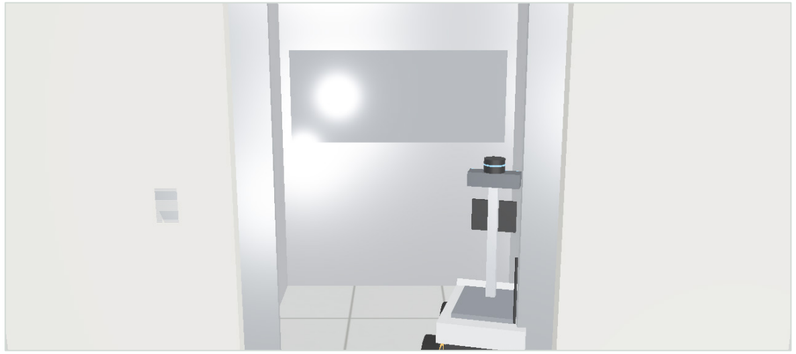
Never cuts through a conversation

left: a normal planner · right: Saathi



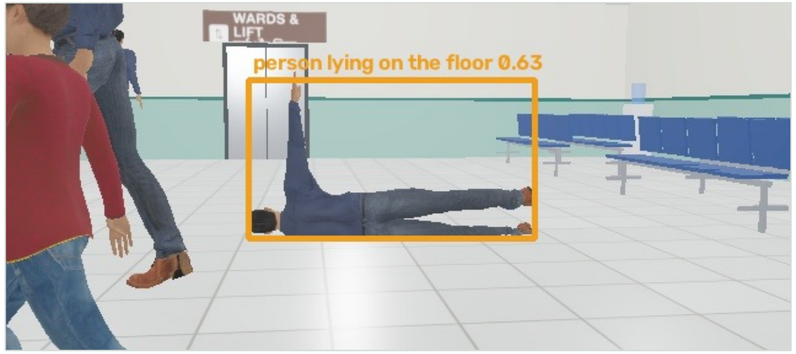
Sees what the map does not have

wet-floor board, cardboard box: 0 hits



Takes the lift to the first floor

doors close, cabin travels, doors open



Spots a fall, alerts the nurse, stays

VLM second opinion: resting or fallen?



Arrives, and says so in Hindi

offline Whisper + phrasebook + Lekha voice

🗣️ **Hindi, fully offline** 📼 **Black box answers “tum ruke kyun?”** 🧠 **Robots share one memory of people** 📊 **Tested on 200 seeded crowds**

Web3 India | builder base | EXIDO

hack in hills

HOW IT WORKS: TWO BRAINS AND A GUARD NEITHER CAN OVERRIDE

SENSORS
Stereo RGB · depth camera · 360° LiDAR · IMU

→

REFLEX BRAIN · 30 Hz
YOLO11 + depth + LiDAR → people with speed · A* on a crowd costmap, 2 s ahead

→

SAFETY GUARD
On LiDAR, below the AI: stop under 0.6 m. Fist = stop.

→

MOTORS
Body velocity → 4 mecanum wheel speeds (physics)

goals ↑ ↓ events

YOU
Hindi voice · typing · hand gesture

→

THINKING BRAIN
Whisper (Hindi) + phrasebook · Qwen3.5-2B VLM describes scenes · Lekha voice

→

SHARED MEMORY
Who each robot saw: where, when, clothes, height. Never a face.

→

BLACK BOX
Every decision + camera frame + reason. Replayable.

ONE REQUEST, END TO END: “BLOOD BANK KAHAN HAI?”

1 Hear
Whisper → “Blood Bank”, first floor

→

2 Plan
A* to the lift lobby on the crowd costmap

→

3 Drive
slow, wait, side-step; guard watches 0.6 m

→

4 Lift
call it, ride up, switch floor map

→

5 Walk
to Blood Bank, around people and groups

→

6 Say
“Aa gaye”; every step in the black box

STACK (ALL ON AN 8 GB M1 LAPTOP, NO CLOUD)

Webots physics	Python 3.10	YOLO11	YOLOE open-vocab	ByteTrack	CLIP
Whisper v3-turbo (MLX)	Qwen3.5-2B VLM (MLX)	A* + social costs	MediaPipe Hands	macOS Lekha TTS	

0.10 m median error placing people

9.5 ms YOLO11n per frame

~5 ms a plan, replanned twice a second

0.3 s fall classified after it happened

PROTOTYPE: ONE RUN, TWO FLOORS, A LIFT, A FALL · 135 s ON ITS OWN CAMERA



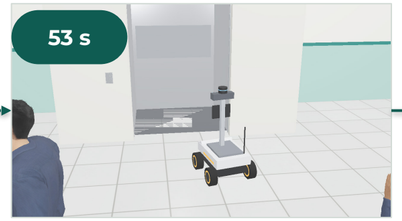
Wet-floor board not on the map: goes around



Arrives at Pharmacy, beside the queue



A man falls: alert sent, Saathi stays



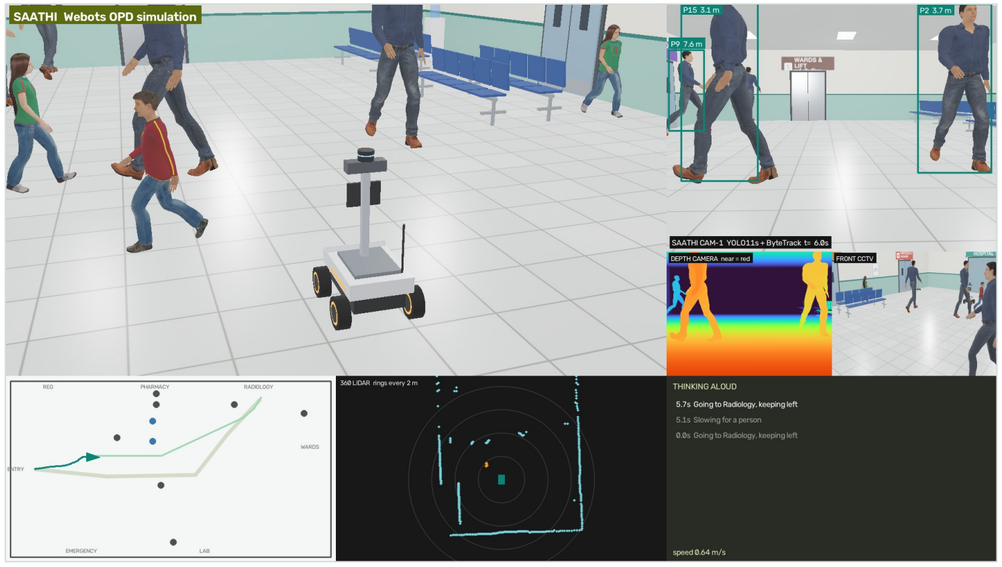
Into the lift; doors close, cabin rises



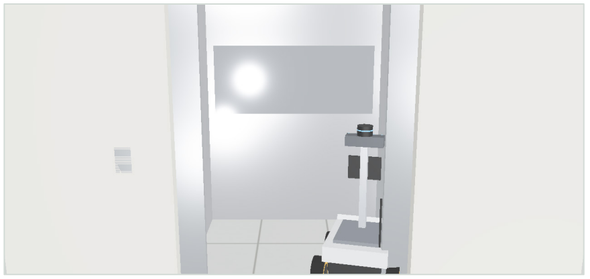
First floor: Nurse Station, Wards, Blood Bank



Back down by lift: arrives at Lab



What Saathi works from: its camera with YOLO boxes and distances, depth, floor map, 360° LiDAR, its thinking log



NEW · the lift

doors, cabin, call button; the map switches floors



NEW · the first floor

Blood Bank, ICU, OT, Wards, nurse station

THIS RUN, MEASURED

0 near-misses

0 conversations cut

0 unmapped obstacles hit

4 departments on 2 floors

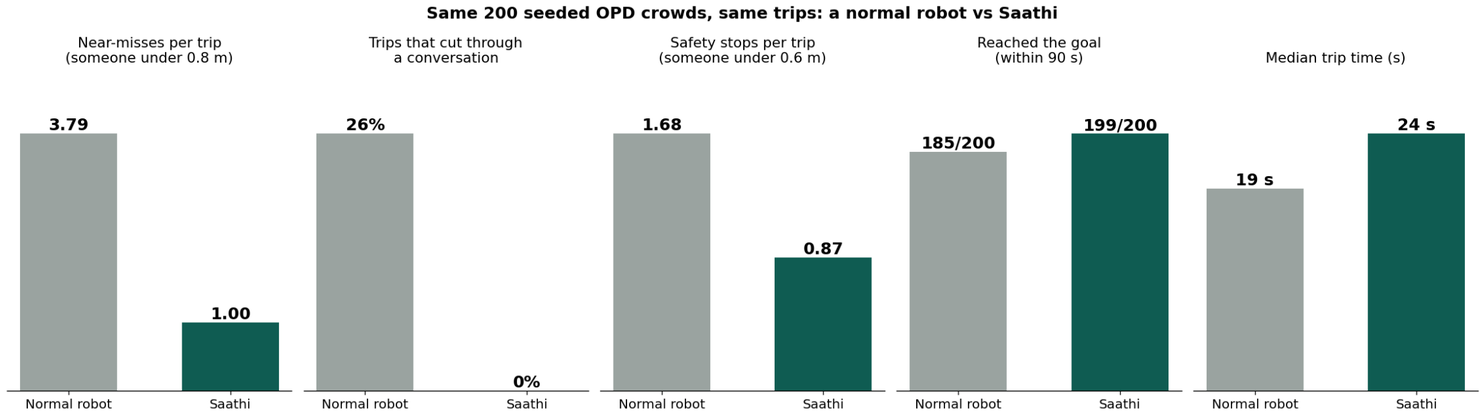
1 fall handled on the way

135 s entrance → Lab, all by itself

NEW IN ROUND 2

- Lift + first floor
- 9 departments by voice
- Tour of departments
- Fall: VLM second opinion
- 2 robots share memory
- Open source on GitHub

RESULTS: 200 SEEDED CROWDS, SAATHI vs A NORMAL ROBOT



NORMAL ROBOT: cuts through the conversation



SAATHI: goes around it



73% fewer

near-misses: 3.79 → 1.00 per trip

0 of 200

trips cut through a conversation (normal robot: 26%)

199/200

trips reached the goal (normal robot: 185/200)

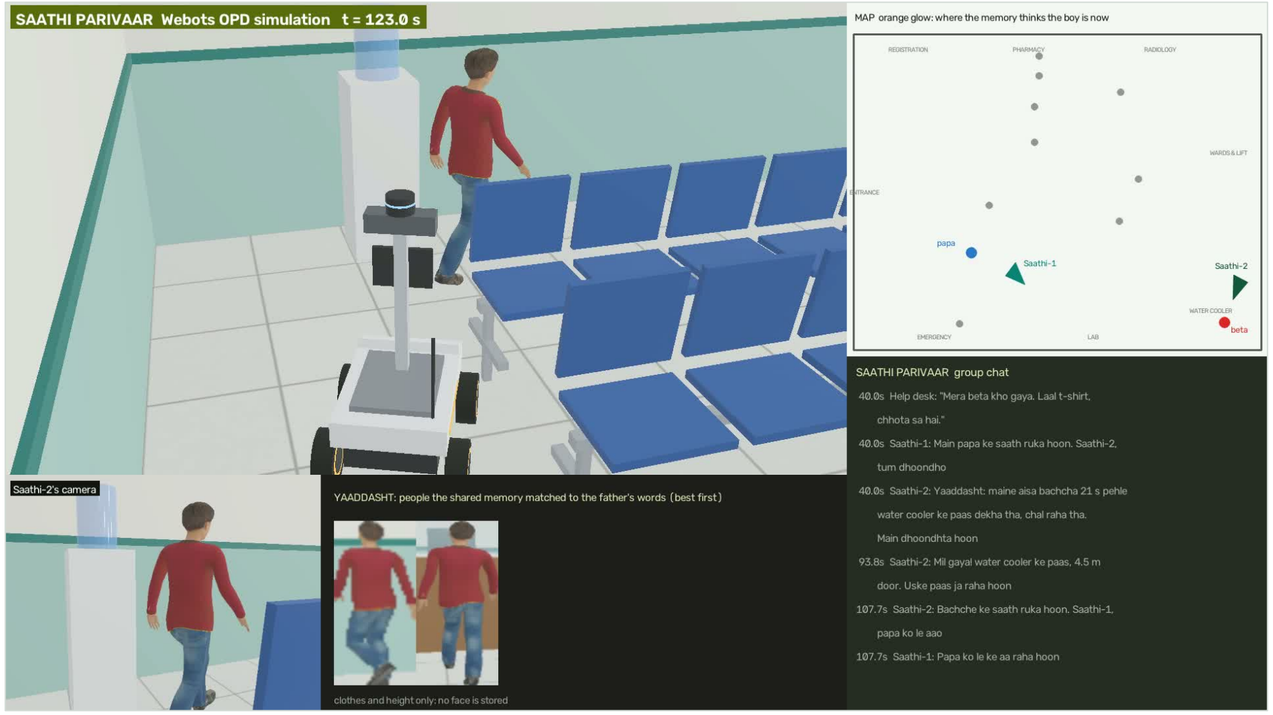
+5 s

the price: median trip 19 s → 24 s

Same crowd, in Webots	Normal	Saathi	Saathi, camera only
Near-misses	4	0	1
Cut a conversation	yes	no	no
Time to Radiology	22 s	25 s	29 s

Each seed: an OPD crowd (8–12 walkers, two talking pairs, a pharmacy queue) and a trip from the entrance to a random department. Normal robot = same A* planner, people as static obstacles, fixed speed, same 0.6 m stop. Deterministic: tools/bench.py 200.

SAATHI PARIVAAR: ROBOTS SHARE ONE MEMORY OF PEOPLE (BUILT, MEASURED)



Two Saathis, one run: papa reports “laal t-shirt” at the help desk; Saathi-1 stays with him, Saathi-2 goes where the memory says the boy is now (orange glow).

1 REMEMBER

who each robot saw: where, when, height, clothes. No face.

2 ASK

papa's words search every robot's memory

3 WHERE NOW?

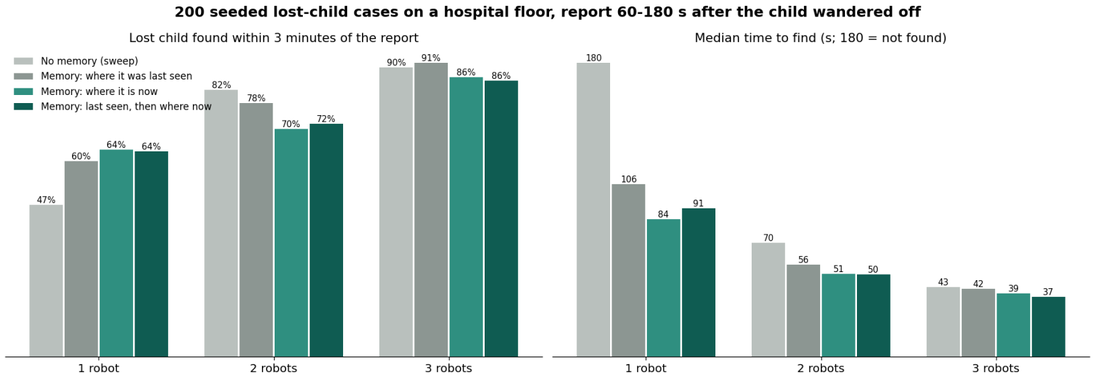
from his last motion; places already checked are ruled out

4 SEARCH AS A TEAM

robots split the places; one stays with papa



Search by description on Saathi's own camera: “small child in a red shirt”, top 8 of 96 people seen are all him. 24 of 24 right.



2.1×

faster with memory, one robot (180 s → 84 s median)

86%

found within 3 min with three robots

200 lost-child seeds, 3 strategies, 1–3 robots (tools/bench_lost.py). Honest: with a team, team size matters more than strategy.

Privacy: clothes, not faces · record deleted after the reunion · wiped daily

PS-01 CHECKLIST WITH PROOF · WHAT IS REAL AND WHAT IS SIMPLIFIED

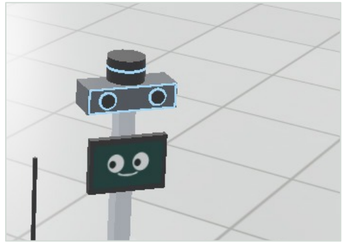
PS-01 asks for	What Saathi does	Proof from our runs
✓ **Understand surroundings with AI vision**	YOLO11 + depth + 360° LiDAR; people on a floor map	People placed within 0.10 m (median)
✓ **Identify relevant people and objects**	Tracks people; YOLOE names objects by text	“cardboard box”, “yellow caution sign” named on sight
✓ **Navigate autonomously to an objective**	A* on a crowd costmap; goals by name, by sight, on either floor	Dustbin found by camera in 12 s · Blood Bank via lift
✓ **Detect and avoid obstacles**	Depth camera finds what the map lacks	Map-only robot hit 2; Saathi 0
✓ **Real-time decision making**	30 Hz reflex loop, replans twice a second	200 crowds: 73% fewer near-misses
✓ **Object detection, image classification**	YOLOE posture: “person lying on the floor”	Fall classified 0.3 s after it happened
✓ **Scene understanding**	Conversations and queues from tracks; signs read	0 of 200 conversations cut
✓ **Vision-language models**	Qwen3.5-2B (Apple MLX) describes the scene	Its description goes with the nurse alert
✓ **Motor control**	Physics mecanum base, speed per wheel	Motion within 7% of the command
✓ **Real or simulated environment**	Two-floor Webots hospital, Hindi + English signs	Deterministic: same seed, same numbers

REAL

- The 3D render; camera, depth and LiDAR data
- YOLO11 / YOLOE / CLIP on the rendered frames
- The planner, the memory, the voice agent
- Physics wheels driven by speed commands; the lift cabin

SIMPLIFIED (WE SAY SO)

- People walk fixed paths and do not react to the robot
- Robot position from the simulator (LiDAR SLAM on the real rover)
- The nurse is simulated (takes over after 6 s)



SAME CODE, REAL ROVER

The brain talks to sensors through one small interface: Webots today, Jetson Orin Nano on our rover next.

PROGRESS: WHAT WAS BUILT WHEN, AND WHAT COMES NEXT

12 OCT ✓
Webots OPD hall, physics rover, crowd-aware planner, camera + LiDAR + depth, fall alert, VLM, black box, 200-crowd A/B

14 OCT ✓
Shared memory of people, lost-child search, 200-seed benchmark, offline Hindi voice

20 OCT ✓
Whole hospital: lift, first floor (Blood Bank, ICU, OT, Wards), tour of departments, GitHub public

24–27 OCT
Demo video, LinkedIn pitch, Instagram reel, Round 2 submission

20–21 NOV
Manali finale: 3 robots live and fully offline on a laptop, held-out 200-seed test

2027
Real rover (Jetson Orin Nano, face screen, mic, e-stop); OPD pilot with consent and face blurring



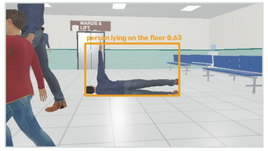
Our rover design: the body Saathi's brain will drive



The first floor, built this week: Blood Bank, Nurse Station, Wards, ICU, OT, stairs, lift







OPEN SOURCE, REPRODUCIBLE

github.com/shaunmortal/saathi

README: GIFs of every run, every number with the script that makes it

- tools/demo.sh: the whole-hospital run, live
- tools/bench.py 200: the A/B numbers
- tools/bench_lost.py 200: the lost-child numbers
- tools/voice.py: talk to it in Hindi

Where it goes: 816 medical colleges, 714 district hospitals; Robot-as-a-Service about ₹15,000 a month per robot

Web3 India | builder base | EXIDO

hack in hills



Thank you!

SAATHI

Raasta bhi, saath bhi.

The way, and the company.

github.com/shaunmortal/saathi

Team Glitch · Ayush Kumar